

Day 13 — Genome Annotation

BIOL 4810

2026-03-08

Day 13 — Genome Annotation + Species Identification

Goal: Use **Prokka** (Galaxy) to annotate a bacterial genome assembly, interpret the output files, extract the **16S rRNA** for a quick BLAST-based ID, and then use **TYGS** + **iTOL** to place your genome on a phylogenomic tree.

Submission: Fill in the answers + upload required screenshots/files below → click **Prepare submission preview** → click **Download as PDF (Print)**.

0) Files you need today

You should already have (from Day 12): - SPAdes assembly contigs file (FASTA)

0A. Assembly file name (as it appears in Galaxy History):

27: SPAdes on dataset 15 and 16 and collection 12: Contigs

Local folder (for Git later)

Create a folder inside your BIOL 4810 course folder, for example: - module2_wgs/day13_prokka_tygs/

0B. Local folder path/description:

Bioinformatics/biol4810/module02/Annotation/

0C) Filter contigs for Prokka (≥ 500 bp)

Some downstream tools struggle with very short contigs or very large numbers of contigs.

In Galaxy: 1. Search for the tool **Filter FASTA** (it may appear as *Filter sequences by length*). 2. Input: your original SPAdes **contigs FASTA**. 3. Set **minimum sequence length = 500 bp**. 4. Run the tool to create a **filtered contigs** dataset.

Use the filtered contigs for Prokka today.

Keep the original (unfiltered) contigs for TYGS. TYGS can take a long time and you already submitted the full assembly there.

0C. Filtered contigs dataset name (Galaxy History):

31: Filter FASTA on dataset 27: FASTA sequences

1) Genome QC (CheckM2 in Galaxy)

Today you will start by evaluating your assembly quality in **Galaxy** using **CheckM2**, which estimates **completeness** and **contamination**.

1A — CheckM2

In Galaxy: 1. Run **CheckM2** on the **filtered assembly FASTA** you will use for Prokka. 2. Open the main CheckM2 results table/report.

CheckM2 tool version (from Job Info):

CheckM2 Galaxy Version 1.1.0+galaxy0

Completeness (%):

98.96

Contamination (%):

0.1

Interpretation (2–3 sentences): Would you trust this assembly for annotation/species ID? Why?

I would trust the assembly for annotation. The completeness is a high number close to 100 and has low contamination value as well.

2) Run Prokka on Galaxy

In Galaxy: 1. Search tools for **Prokka** 2. Input: your SPAdes **contigs FASTA** 3. Use defaults unless instructed otherwise 4. Run Prokka

Record your Prokka settings

2A. Prokka version (from Job Info):

Prokka Galaxy Version 1.14.6+galaxy1

2B. Any settings changed from defaults? If yes, what?

Force GenBank compliance- YES

3) Prokka outputs — what is each file for?

From Prokka outputs in Galaxy, you will use these most today:

- **.txt** summary statistics
- **.gff** master annotation (features + coordinates)

- **.ffn** nucleotide sequences of gene features (includes rRNA genes)
- **.faa** amino acid sequences of predicted proteins

3A. In your own words, what is the difference between .gff , .ffn , and .faa ? (2–4 sentences)

.gff is the master annotation which has all the features of the annotation and coordinates. .ffn us the nucleiptide sequence of the genes and it contains the rRNA genes. .faa is the amino acid sequences of proteins, this contains hypothetical proteins.

4) Interpret the Prokka summary (.txt)

Open the Prokka **.txt** summary.

4A. Organism line (paste exactly as reported — it may be “Unknown”):

Genus species strain

4B. contigs:

79

4C. bases:

4797060

4D. CDS:

4507

3E. rRNA (total rRNA features):

3

3F. tRNA:

62

3G. tmRNA:

1

3H. Based on genome size and CDS count, does this look like a typical bacterial genome? Why/why not? (2–3 sentences)

Yes it does look like a typical bacterial genome. It is in the range of the amount of bases that is in a typical bacterial genome (3-5 mbp). It is also in range of the number of CDS that is in a typical bacterial genome (3,000- 5,000).

5) rRNA features (16S/23S/5S) — paste evidence from Prokka

The Prokka .txt summary often reports only a total count (e.g., **rRNA: 3**) without listing 16S vs 23S vs 5S in that summary.

To see which rRNAs were found: - Open the .gff file and search for rRNA , 16S , 23S , 5S **OR** - Open the .ffn file and search for ribosomal RNA

5A. Paste the description line (header or attribute text) for the 16S rRNA feature:

```
gnl|Prokka|NCJPFIPA_56   barrnap:0.9   rRNA    51  1588    0   -   .  
ID=NCJPFIPA_04563;Parent=NCJPFIPA_04563_gene;locus_tag=NCJPFIPA_04563;product=16S ribosomal
```

5B. Paste the description line (header or attribute text) for the 23S rRNA feature:

```
ID=NCJPFIPA_04556;Parent=NCJPFIPA_04556_gene;locus_tag=NCJPFIPA_04556;note=aligned only 53  
percent of the 23S ribosomal RNA;product=23S ribosomal RNA (partial)
```

5C. Paste the description line (header or attribute text) for the 5S rRNA feature:

```
RNAgnl|Prokka|NCJPFIPA_8   barrnap:0.9   rRNA    252  362  7.3e-11   +   .  
ID=NCJPFIPA_02254;Parent=NCJPFIPA_02254_gene;locus_tag=NCJPFIPA_02254;product=5S ribosomal
```

6) Hypothetical proteins — estimate and interpret

Hypothetical proteins typically appear as: - product=hypothetical protein in .gff - “hypothetical protein” in headers/annotations in .faa

6A. How did you estimate or count hypothetical proteins? (1–2 sentences)

Command f and typed in hypothetical

6B. Estimated number of hypothetical proteins:

1098

6C. Estimated % hypothetical proteins (of CDS):

24.01%

6D. What does a high % hypothetical proteins mean (and NOT mean)? (2–3 sentences)

A high percentage of hypothetical proteins means that a large portion of a newly sequenced genome lack experimental evidence of expression or known homologs in current databases. It does not mean that these sequences are junk DNA or junk proteins. It also doesn't mean that they are non-functional.

7) Extract 16S rRNA from the Prokka .ffn

Open the Prokka .ffn file. Search for a header/annotation indicating **16S ribosomal RNA**.

7A. Paste the FASTA header line for the 16S entry you used:

```
>NCJPFIPA_04563 16S ribosomal RNA
```

7B. Paste the full 16S rRNA sequence you used for BLAST (sequence only; you can keep line breaks):

```
ATGCAACGCGAAGAACCTTACCTGGTCTTGACATCCACAGAAGAATCCAGAGATGGATTT
GTGCCTTCGGGAAGTGTGAGACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTTGTGAAAT
GTTGGGTTAAGTCCCGCAACGAGCGCAACCCTTATCCTTTGTTGCCAGCGATTAGGTCGG
GAACTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATC
ATGGCCCTTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACC
TCGCGAGAGCAAGCGGACCTCATAAAGTGCCTCGTAGTCCGGATTGGAGTCTGCAACTCG
ACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCC
CGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAAAGAAGTAGGTAGC
TTAACCTTCGGGAGGGCGCTTACCACTTTGTGATTCATGACTGGGGTGAAGTCGTAACAA
GGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTT
```

7C. In 1 sentence: why do we use 16S for a quick ID?

We used 16S for rapid bacterial identification because it is universally present in all bacteria, and it contains a highly conserved region.

8) 16S BLAST (species ID quick check)

BLAST your 16S sequence: 1. Go to NCBI BLAST 2. Choose **BLASTn** 3. Paste your 16S sequence 4. Run BLAST and choose a reasonable hit to interpret

Upload screenshot: your 16S BLAST results (Descriptions table or Alignment view)

Choose File

Screenshot...astresults

BLAST® » blastn suite » results for RID-V0R09DB8016

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Job Title Nucleotide Sequence

RID [V0R09DB8016](#) Search expires on 03-12 01:59 am [Download All](#)

Program BLASTN [Citation](#)

Database rRNA_typestrains/16S_ribosomal_RNA [See details](#)

Query ID lcl|Query_2465289

Description None

Molecule type dna

Query Length 1538

Other reports [Distance tree of results](#) [MSA viewer](#)

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

[Add organism](#)

Percent Identity to **E value** to **Query Coverage** to

[Filter](#) [Reset](#)

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments Download Select columns Show 100

☒ select all 100 sequences selected

[GenBank](#) [Graphics](#) [Distance tree of results](#) [MSA Viewer](#)

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ Salmonella enterica subsp. enterica serovar Typhimurium strain LT2 16S ribosomal RNA, partial sequence	Salmonella e...	2784	2784	100%	0.0	99.48%	1544	NR_074910.1
☒ Salmonella enterica subsp. enterica serovar Typhimurium strain ATCC 13311 16S ribosomal RNA, partial sequence	Salmonella e...	2778	2778	100%	0.0	99.15%	1539	NR_119108.1
☒ Salmonella bongori strain NCTC 12419 16S ribosomal RNA, complete sequence	Salmonella b...	2741	2741	100%	0.0	98.83%	1542	NR_074888.1
☒ Citrobacter telavivensis strain 6105 16S ribosomal RNA, complete sequence	Citrobacter t...	2691	2691	100%	0.0	98.24%	1542	NR_180890.1
☒ Salmonella enterica subsp. enterica strain LT2 16S ribosomal RNA, partial sequence	Salmonella e...	2689	2689	95%	0.0	99.66%	1467	NR_104709.1

[Feedback](#)

8A. Top hit organism name (as shown):

Salmonella enterica subsp. enterica serovar Typhimurium strain LT2 16S ribosomal RNA, partial sequence

8B. % identity and query coverage for that hit:

99.48% identities and 100% Query Cover

8C. Your best interpretation of species/genus (be cautious) (2–3 sentences):

The hit we got was Salmonella typhimurium. The query cover is 100 and the e-value is 0.0. This means that it is basically a direct hit for this species.

8D. One reason 16S BLAST can be misleading or insufficient (1–2 sentences):

16S BLAST can be misleading because 16S is a single gene. This gene can be in multiple bacteria not just S. typhimurium.

9) ANI confirmation (EZBioCloud ANI)

After you have a likely species ID from **16S BLAST**, use ANI as a stronger genome-wide comparison.

You will use the **EZBioCloud ANI** tool here (open in a new tab): <https://www.ezbiocloud.net/tools/ani>
(<https://www.ezbiocloud.net/tools/ani>)

What you will do 1. Download your **FILTERED contigs FASTA** (the same one you used for Prokka today). 2. Download **two reference genomes** (FASTA) from NCBI for *two candidate species* you want to compare against (often the top 16S BLAST hit and one close alternative). 3. Run ANI twice (your genome vs reference genome 1; your genome vs reference genome 2).

9A. Species/strain for comparison #1 (your chosen reference):

salmonella typhimurium

9B. ANI for comparison #1 (%):

99.94%

9C. Species/strain for comparison #2 (your chosen reference):

Salmonella Bongori

9D. ANI for comparison #2 (%):

89.82%

9E. Interpretation (2–3 sentences): Which comparison supports the strongest species-level match, and why?

The salmonella typhimurium comparison supports the strongest species-level match. It has a higher average nucleotide identity than the species Salmonella Bongori. S. typhimurium has a 99.94% ANI, while S. Bongori has a 89.82% ANI.

10) TYGS — what it provides

Submit your assembly to TYGS (Type Strain Genome Server). TYGS will: - Compare your genome to type strains - Provide **dDDH** (species boundary ~70%) - Produce a phylogenomic tree

10A. In your own words: what is TYGS doing that 16S BLAST is not? (2–4 sentences)

TYGS compares entire genomes to determine how closely related an organism is to known species. BLAST with the 16S rRNA gene only compares one gene, so it is less precise for identifying closely related species.

10B. Record your closest type strain match (name) and dDDH value (if available today):

Salmonella typhimurium JCM 1652

10C. What does the 70% dDDH threshold mean? (1–2 sentences)

The 70% digital dDDH threshold means that if two organisms have 70% or more genome similarity, they are considered the same species. If the dDDH value is below 70%, the organisms are considered different

10D. If TYGS shows low branch support and offers a proteome-based tree, what does that imply? (2–3 sentences)

If TYGS shows low branch support, it means the evolutionary relationship between those organisms is uncertain or weakly supported by the data. When it offers a proteome-based tree, it implies that comparing all proteins in the genome may provide a clearer and more reliable phylogenetic relationship than the original

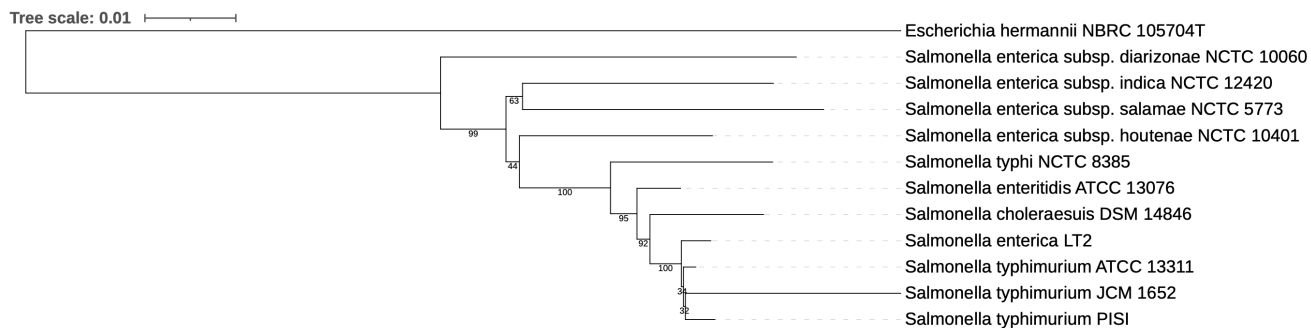
11) iTOL — visualize your TYGS tree

1. From TYGS, download the **tree file (.phy)**
2. Go to **iTOL** (Interactive Tree of Life)
3. Upload the tree
4. Adjust labels/formatting enough to be readable
5. Take a screenshot

Upload screenshot: iTOL tree visualization (from TYGS tree)

Choose File

Screensho....54.01 AM



11A. What does the tree tell you that a single BLAST hit does not? (2–3 sentences)

A phylogenetic tree shows the evolutionary relationships among many organisms at once and how closely they are related to each other. A single BLAST hit only shows the closest sequence match, but it does not show the overall evolutionary context or relationships between multiple species.

11B. Identify one branch you trust and one you would be cautious about. Explain using support values if shown. (2–4 sentences)

One branch I trust is the grouping of *Salmonella typhimurium* strains, which has a support value of 100, indicating very strong confidence in that relationship. A branch I would be cautious about is one with a low support value like 32, because this suggests weak evidence for that evolutionary relationship and the placement could change with more data.

12) What to download and save for Git

Minimum recommended downloads for your local folder: - Prokka **.txt** summary - Prokka **.gff** - Prokka **.ffn** - Prokka **.faa** - CheckM2 results/report (table) - TYGS tree file (**.phy**) - Your PDF from this page (print-to-PDF)

12A. Write 4–6 lines of a “workflow note” you would put in `workflow_day13.md` (include tool versions when possible).

-Genome annotation done with Prokka Galaxy Version 1.14.6+galaxy1 with parameters of "Force GenBank compliance- YES." Species ID done with BLASTn (web) with parameters of "rRNA/ITS database."
-Build phylogenetic tree with Type Strain Genome Server with parameters of "Whole Proteome Tree."
-Visualize phylogenetic tree with Interactive Tree of Life v7 (web) with parameters of "Display bootstraps as text" -Average nucleotide identity with <https://www.ezbiocloud.net/tools/ani> with parameters of "From ncbi genome database (<https://www.ncbi.nlm.nih.gov/datasets/genome/>), downloaded the FASTA file for

Export (PDF)

1. Upload the required screenshots above (BLAST + iTOL)
2. Click **Prepare submission preview**
3. Click **Download as PDF (Print)**

Submission preview

Setup

Assembly file: 27: SPAdes on dataset 15 and 16 and collection 12: Contigs

Local folder: Bioinformatics/biol4810/module02/Annotation/

Prokka run

Version: Prokka Galaxy Version 1.14.6+galaxy1

Settings: Force GenBank compliance- YES

Outputs understanding

.gff vs *.ffn* vs *.faa*: *.gff* is the master annotation which has all the features of the annotation and coordinates. *.ffn* is the nucleotide sequence of the genes and it contains the rRNA genes. *.faa* is the amino acid sequences of proteins, this contains hypothetical proteins.

Prokka summary (.txt)

Organism: Genus species strain

Contigs: 79 *Bases*: 4797060

CDS: 4507 *rRNA*: 3 *tRNA*: 62 *tmRNA*: 1

Sanity check: Yes it does look like a typical bacterial genome. It is in the range of the amount of bases that is in a typical bacterial genome (3-5 mbp). It is also in range of the number of CDS that is in a typical bacterial genome (3,000- 5,000).

rRNA evidence

16S description: gn|Prokka|NCJPFIPA_56 barrnap:0.9 rRNA 51 1588 0 - .

ID=NCJPFIPA_04563;Parent=NCJPFIPA_04563_gene;locus_tag=NCJPFIPA_04563;product=16S ribosomal

23S description: gn|Prokka|NCJPFIPA_53 barrnap:0.9 rRNA 4 1724 0 + .

ID=NCJPFIPA_04556;Parent=NCJPFIPA_04556_gene;locus_tag=NCJPFIPA_04556;note=aligned only 53 percent of the 23S ribosomal RNA;product=23S ribosomal RNA (partial)

5S description: RNAgn|Prokka|NCJPFIPA_8 barrnap:0.9 rRNA 252 362 7.3e-11 + .

ID=NCJPFIPA_02254;Parent=NCJPFIPA_02254_gene;locus_tag=NCJPFIPA_02254;product=5S ribosomal RNA

CheckM2

CheckM2 version: CheckM2 Galaxy Version 1.1.0+galaxy0

Completeness: 98.96% *Contamination*: 0.1%

CheckM2 interpretation: I would trust the assembly for annotation. The completeness is a high number close to 100 and has low contamination value as well.

Hypothetical proteins

How estimated: Command *f* and typed in hypothetical

hypothetical: 1098 % *hypothetical*: 24.01%

Interpretation: A high percentage of hypothetical proteins means that a large portion of a newly sequenced genome lack experimental evidence of expression or known homologs in current databases. It does not mean that these sequences are junk DNA or junk proteins. It also doesn't mean that they are non-functional.

16S extraction

16S header: >NCJPFIPA_04563 16S ribosomal RNA

16S sequence:

```
TTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTC
GAACGGTAACGGGAAGCAGCTTGCTGCTTCGCTGACGAGTGGCGGACGGGTGAGTAATGT
CTGGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTGGCTAATACCGCATAAC
GTCGCAAGACCAAAGAGGGGGACCTTCGGGCCTCTTGCCATCAGATGTGCCCAGATGGGA
TTAGCTTGTTGGTGAGGTAACGGCTCACCAAGGCGACGATCCCTAGCTGGTCTGAGAGGA
TGACCAGCCACACTGGAAGTGAAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGA
ATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCTTCG
GGTTGTAAAGTACTTTCAGCGGGGAGGAAGGTGTTGTGGTTAATAACCGCAGCAATTGAC
GTTACCCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGT
GCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCGGAT
GTGAAATCCCCGGGCTCAACCTGGGAACTGCATTCGAAACTGGCAGGCTTGAGTCTTGTA
GAGGGGGGTAGAATTCCAGGTGTAGCGGTGAAATGCGTAGAGATCTGGAGGAATACCGGT
GGCGAAGGCGGCCCCCTGGACAAAGACTGACGCTCAGGTGCGAAAGCGTGGGGAGCAAAC
AGGATTAGATAACCTGGTAGTCCACGCCGTAAACGATGTCTACTTGGAGGTTGTGCCCTT
GAGGCGTGCTTCCGGAGCTAACGCGTTAAGTAGACCGCCTGGGGAGTACGGCCGCAAGG
TTAAAACTCAAATGAATTGACGGGGGCCCCGCACAAGCGGTGGAGCATGTGGTTTAATTCG
ATGCAACGCGAAGAACCCTTACCTGGTCTTGACATCCACAGAAGAATCCAGAGATGGATTT
GTGCCTTCGGGAACTGTGAGACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTGTGAAAT
GTTGGGTAAAGTCCCGCAACGAGCGCAACCCTTATCCTTTGTTGCCAGCGATTAGGTCCG
GAACTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATC
ATGGCCCTTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACC
TCGCGAGAGCAAGCGGACCTCATAAAGTGCGTCGTAGTCCGGATTGGAGTCTGCAACTCG
ACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCC
CGGGCCTTGTTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAAGAAGTAGGTAGC
TTAACCTTCGGGAGGGCGCTTACCACCTTGTGATTTCATGACTGGGGTGAAGTCGTAACAA
GGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTT
```

Why 16S: We used 16S for rapid bacterial identification because it is universally present in all bacteria, and it contains a highly conserved region.

16S BLAST

Top hit: Salmonella enterica subsp. enterica serovar Typhimurium strain LT2 16S ribosomal RNA, partial sequence

Metrics: 99.48% identities and 100% Query Cover

Interpretation: The hit we got was Salmonella typhimurium. The query cover is 100 and the e-value is 0.0. This means that it is basically a direct hit for this species.

Limitation: 16S BLAST can be misleading because 16S is a single gene. This gene can be in multiple bacteria not just S. typhimurium.

BLAST screenshot

BLAST® » blastn suite » results for RID-V0R09DB8016

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Job Title Nucleotide Sequence

RID V0R09DB8016 Search expires on 03-12 01:59 am [Download All](#)

Program BLASTN [Citation](#)

Database rRNA_typerstrains/16S_ribosomal_RNA [See details](#)

Query ID lcl|Query_2465289

Description None

Molecule type dna

Query Length 1538

Other reports [Distance tree of results](#) [MSA viewer](#)

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

[Add organism](#)

Percent Identity to **E value** to **Query Coverage** to

[Filter](#) [Reset](#)

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments [Download](#) [Select columns](#) Show 100

☒ select all 100 sequences selected [GenBank](#) [Graphics](#) [Distance tree of results](#) [MSA Viewer](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Salmonella enterica subsp. enterica serovar Typhimurium strain LT2 16S ribosomal RNA, p...	Salmonella e...	2784	2784	100%	0.0	99.48%	1544	NR_074910.1
☒	Salmonella enterica subsp. enterica serovar Typhimurium strain ATCC 13311 16S ribosoma...	Salmonella e...	2778	2778	100%	0.0	99.15%	1539	NR_119108.1
☒	Salmonella bongori strain NCTC 12419 16S ribosomal RNA, complete sequence	Salmonella b...	2741	2741	100%	0.0	98.83%	1542	NR_074888.1
☒	Citrobacter telavivensis strain 6105 16S ribosomal RNA, complete sequence	Citrobacter t...	2691	2691	100%	0.0	98.24%	1542	NR_180890.1
☒	Salmonella enterica subsp. enterica strain LT2 16S ribosomal RNA, partial sequence	Salmonella e...	2689	2689	95%	0.0	99.66%	1467	NR_104709

TYGS

TYGS vs 16S: TYGS compares entire genomes to determine how closely related an organism is to known species. BLAST with the 16S rRNA gene only compares one gene, so it is less precise for identifying closely related species.

Closest match + dDDH: Salmonella typhimurium JCM 1652

70% threshold: The 70% digital dDDH threshold means that if two organisms have 70% or more genome similarity, they are considered the same species. If the dDDH value is below 70%, the organisms are considered different species.

Proteome tree option (low support): If TYGS shows low branch support, it means the evolutionary relationship between those organisms is uncertain or weakly supported by the data. When it offers a proteome-based tree, it implies that comparing all proteins in the genome may provide a clearer and more reliable phylogenetic relationship than the original analysis.

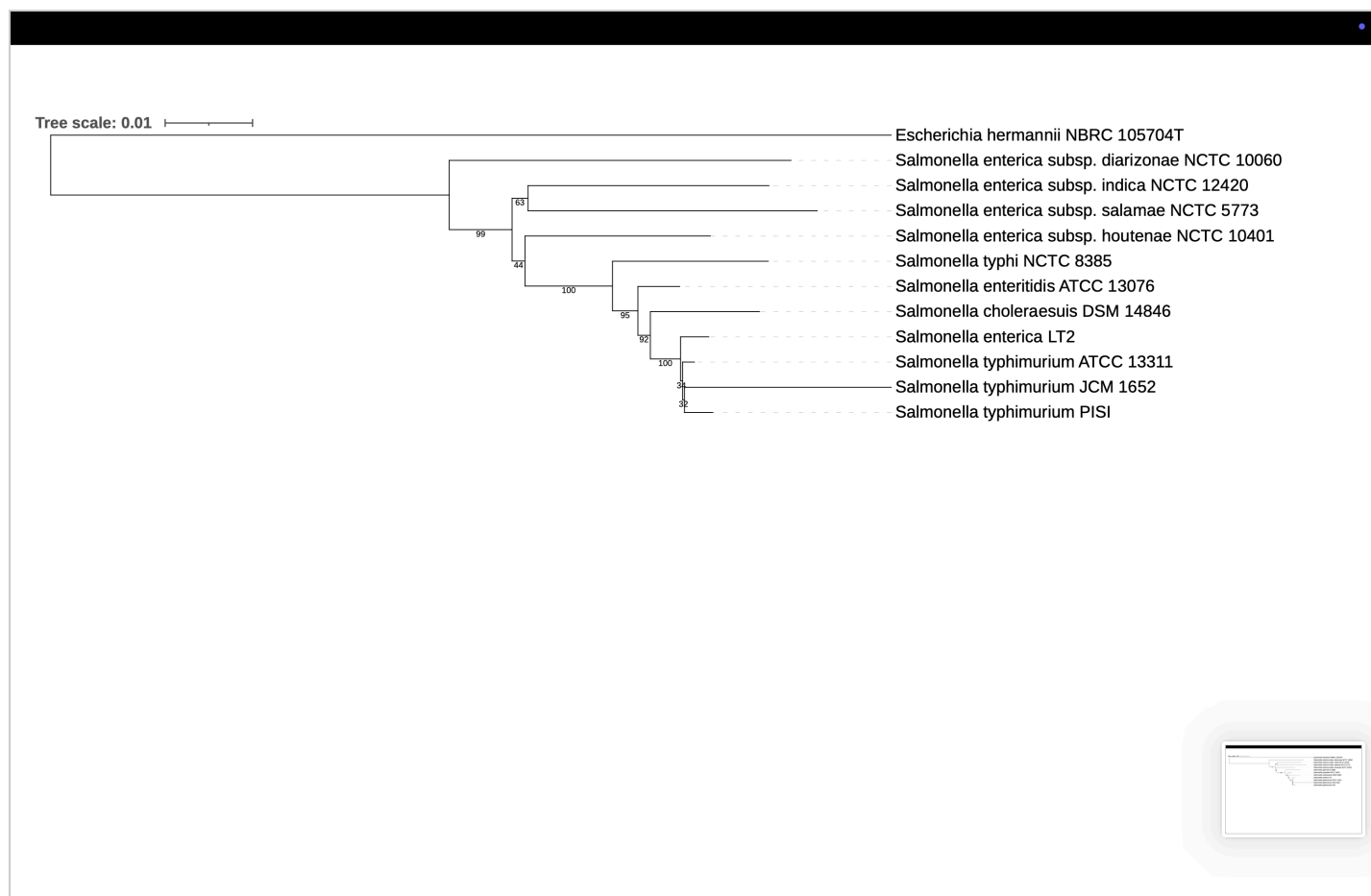
iTOL tree

Tree adds value because: A phylogenetic tree shows the evolutionary relationships among many organisms at once and how closely they are related to each other. A single BLAST hit only shows the closest sequence match, but it does not show the overall evolutionary context or relationships between multiple species.

Support/trust: One branch I trust is the grouping of Salmonella typhimurium strains, which has a support value of 100, indicating very strong confidence in that relationship. A branch I would be cautious about is one with a low support value like 32, because this suggests weak evidence for that evolutionary

relationship and the placement could change with more data.

iTOL tree screenshot



Workflow note (for Git)

- Filter FASTA on Galaxy Version 2.3 with parameters of "Filtered on sequence length; minimum length 500 bp."
- Quality assessment using CheckM2 Galaxy Version 1.1.0+galaxy0 with parameters "Model options- Output Quality predictions for both models for each genome."
- Genome annotation done with Prokka Galaxy Version 1.14.6+galaxy1 with parameters of "Force GenBank compliance- YES." Species ID done with BLASTn (web) with parameters of "rRNA/ITS database."
- Build phylogenetic tree with Type Strain Genome Server with parameters of "Whole Proteome Tree."
- Visualize phylogenetic tree with Interactive Tree of Life v7 (web) with parameters of "Display bootstraps as text" -Average nucleotide identity with <https://www.ezbiocloud.net/tools/ani> with parameters of "From ncbi genome database (<https://www.ncbi.nlm.nih.gov/datasets/genome/>), downloaded the FASTA file for Salmonella Typhimurium (closest hit on 16S BLAST) and Salmonella bongori (more distant relative). Compared filteredcontigs.fasta to the FASTA files from Typhimurium and S. Bongori."